



National Academy of science and Technology

[Affiliated to Pokhara University]

Dhangadhi, Kailali (Nepal)

Accredited by University Grants Commission (UGC), Nepal (2022)

Answer Book No. 1958

Nos. of Additional answer books:-

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1. The student must write the answers on the both sides of the answer book.
2. A margin of one inch should be left on each page.
3. The signature of the invigilator is essential in each sheet.
4. If required loose paper will be provided for scribing and no paper is to be brought in the exam hall for this purpose. Any student found with any loose or other (except admission card) in his/her possession will be expelled from the examinations. All works must be done on the copies provided by the college and pages should not be torn out.
5. It is prohibited to write anythings else on the question paper or elsewhere except at answer book.
6. The student must check the used answer books before delivery. Additional answer sheet must be properly stitched.
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MARKS OBTAINED				
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Q.No.	A	B	C	Total
1				
2				
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TOTAL OBTAINED				

Information to be filled by the student	
Name :	Prashant Bhatta
Programme :	BE Computer
Semester :	III
Level/Grade :	Bachelor
Section :	
Roll No. Class :	34
Exam Roll No. :	22070226
Faculty :	Science and Technology
Examination :	Preboard
Subject :	Calculus - II
Date :	2080-10-24

To be used only by the college/School authority	
Remarks by the exam authority (If any):	
i)	
ii)	61
iii)	
iv)	

Signature of the Invigilator : i)  ii)

Signature of the Examiner

Signature of the Controller

Q. no. 1 (b)

Dirichlet's theorem states that if integral is in the form

$$\iiint_V x^{l-1} y^{m-1} z^{n-1} dV$$

then, it's volume can be defined as

$$V = \frac{l! m! n!}{(l+m+n+1)!}$$

Now, triple integral can be calculated as,

$$\iiint_V x^2 dx dy dz$$

$$= \iiint_V x^{3-1} y^{1-1} z^{1-1} dx dy dz$$

$$= \frac{3! 1! 1!}{(3+1+1+1)!}$$

$$\begin{aligned}
 &= \frac{2! \cdot 0! \cdot 0!}{5!} \\
 &= \frac{2}{5!} \\
 &= \frac{2}{5 \times 4 \times 3 \times 2 \times 1} \\
 &= \frac{1}{60}
 \end{aligned}$$

Q. no. 2. (a)

Given,

$$(1-x^2)y'' - 2xy' + 2y = 0 \quad \text{--- (i)}$$

$$\begin{aligned}
 y &= a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 \\
 &\quad + a_5x^5 + a_6x^6 + a_7x^7 + \dots \quad \text{--- (ii)}
 \end{aligned}$$

$$\begin{aligned}
 y' &= a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 \\
 &\quad + 5a_5x^4 + 6a_6x^5 + 7a_7x^6 + \dots \quad \text{--- (iii)}
 \end{aligned}$$

$$\begin{aligned}
 y'' &= 2a_2 + 6a_3x + 12a_4x^2 + 20a_5x^3 \\
 &\quad + 30a_6x^4 + 42a_7x^5 + \dots \quad \text{--- (iv)}
 \end{aligned}$$

Then, from

$$(1-x^2)(a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + a_6x^6 + a_7x^7 + \dots)$$

$$- 2x(a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + 5a_5x^4 + 6a_6x^5 + 7a_7x^6 + \dots)$$

$$+ 2(a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + a_6x^6 + a_7x^7 + \dots)$$

$$(1-x^2)(2a_2 + 6a_3x + 12a_4x^2 + 20a_5x^3 + 30a_6x^4 + 42a_7x^5 + \dots)$$

$$- 2x(2a_1 + 4a_2x + 6a_3x^2 + 8a_4x^3 + 10a_5x^4 + 12a_6x^5 + 14a_7x^6 + \dots)$$

$$+ 2(a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + a_6x^6 + a_7x^7 + \dots)$$

$$\text{or, } 2a_2 + 6a_3x + 12a_4x^2 + 20a_5x^3 + 30a_6x^4 + 42a_7x^5 + \dots$$

$$- 2a_1x - 4a_2x^2 - 6a_3x^3 - 8a_4x^4 - 10a_5x^5 - 12a_6x^6 - 14a_7x^7 - \dots$$

$$+ 2a_0 + 2a_1x + 2a_2x^2 + 2a_3x^3 + 2a_4x^4 + 2a_5x^5 + 2a_6x^6 + 2a_7x^7 + \dots$$

$$+ 2a_2x^2 + 6a_3x^3 + 12a_4x^4 + 20a_5x^5 + 30a_6x^6 + 42a_7x^7 + \dots$$

$$\text{or, } (2a_2 + 6a_3x + 12a_4x^2 + 20a_5x^3 + 30a_6x^4 + 42a_7x^5 + \dots)$$

$$- (2a_1x + 4a_2x^2 + 6a_3x^3 + 8a_4x^4 + 10a_5x^5 + 12a_6x^6 + 14a_7x^7 + \dots)$$

Then, from ①,

$$(1-x^2)(a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + a_6x^6 + a_7x^7 + \dots)$$

$$= 2x(a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + 5a_5x^4 + 6a_6x^5 + 7a_7x^6 + \dots)$$

$$+ 2$$

$$(1-x^2)(2a_2 + 6a_3x + 12a_4x^2 + 20a_5x^3 + 30a_6x^4 + 42a_7x^5 + \dots)$$

$$- 2x(a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + 5a_5x^4 + 6a_6x^5 + 7a_7x^6 + \dots)$$

$$+ 2(a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + a_6x^6 + a_7x^7 + \dots) = 0$$

$$\text{or, } (2a_2 + 6a_3x + 12a_4x^2 + 20a_5x^3 + 30a_6x^4 + 42a_7x^5 + \dots)$$

$$+ (-2a_2x^2 - 6a_3x^3 - 12a_4x^4 - 20a_5x^5 - 30a_6x^6 - 42a_7x^7 - \dots)$$

$$+ (-2a_1x - 4a_2x^2 - 6a_3x^3 - 8a_4x^4 - 10a_5x^5 - 12a_6x^6 - 14a_7x^7 - \dots)$$

$$+ (2a_0 + 2a_1x + 2a_2x^2 + 2a_3x^3 + 2a_4x^4 + 2a_5x^5 + 2a_6x^6 + 2a_7x^7 + \dots) = 0$$

$$\text{or, } (2a_2 + 2a_0) + (6a_3 - 2a_1 + 2a_1)x$$

$$+ (12a_4 - 2a_2 - 4a_2 + 2a_2)x^2$$

$$+ (20a_5 - 6a_3 - 6a_3 + 2a_3)x^3 +$$

$$(30a_6 - 12a_4 - 8a_4 + 2a_4)x^4 + (42a_7 - 20a_5 - 10a_5 + 2a_5)x^5 + \dots = 0$$

Comparing like terms on both sides, we get

$$2a_2 + 2a_0 = 0$$

or, $a_2 = -a_0$

$$\text{Also, } 6a_3 - 2a_1 + 2a_1 = 0$$

$$\text{or, } 6a_3 = 0$$

$$\text{or, } a_3 = 0$$

$$\text{Also, } 12a_4 - 2a_2 - 4a_2 + 2a_2 = 0$$

$$\text{or, } 12a_4 - 4a_2 = 0$$

$$\text{or, } 12a_4 = 4a_2$$

$$\therefore a_4 = \frac{1}{3} a_2 = -\frac{1}{3} a_0$$

$$\text{Also, } 20a_5 - 6a_3 - 6a_3 + 2a_3 = 0$$

$$\text{or, } 20a_5 = 0$$

$$\therefore a_5 = 0 \quad (\because a_3 = 0)$$

$$\text{Also, } 30a_6 - 12a_4 - 8a_4 + 2a_4 = 0$$

$$\text{or, } 30a_6 - 18a_4 = 0$$

0

$$\text{or, } 30a_6 = 18a_4$$

$$\text{or, } a_6 = \frac{18}{30} a_4$$

ides

$$\begin{aligned}\text{or, } a_6 &= \frac{3}{5} a_4 = \frac{3}{5} \left(-\frac{1}{3} a_0 \right) \\ &= -\frac{1}{5} a_0\end{aligned}$$

Then, from ①,

$$y = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + a_5 x^5 + a_6 x^6 + \dots$$

$$= a_0 + a_1 x - a_0 x^2 + 0 - \frac{1}{3} a_0 x^4$$

$$+ 0 - \frac{1}{5} a_0 x^6 + \dots$$

$$= a_1 x + a_0 \left(1 - x^2 - \frac{1}{3} x^4 - \frac{1}{5} x^6 - \dots \right)$$

⑧

Q. no. 2 (b)

We know that

$$J_n(x) = \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m+n}}{2^{2m+n} \cdot m! \cdot (m+n)!}$$

L. H. S.

$$= \frac{d}{dx} [x^{-n} J_n(x)]$$

$$= \frac{d}{dx} \left[x^{-n} \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m+n}}{2^{2m+n} \cdot m! \cdot (m+n)!} \right]$$

$$= \frac{d}{dx} \left[\sum_{m=0}^{\infty} \frac{(-1)^m x^{2m+n-n}}{2^{2m+n} \cdot m! \cdot (m+n)!} \right]$$

$$= \frac{d}{dx} \left[\sum_{m=0}^{\infty} \frac{(-1)^m x^{2m}}{2^{2m+n} \cdot m! \cdot (m+n)!} \right]$$

$$= \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m+n}}{2^{2m+n} \cdot m! \cdot (m+n)!}$$

$$= \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m+n}}{2^{2m+n} \cdot m! \cdot (m+n)!}$$

$$= \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m+n}}{2^{2m+n} \cdot m! \cdot (m+n)!}$$

$$= x^{-n}$$

$$P$$

$$=$$

$$=$$

$$=$$

$$= \sum_{m=0}^{\infty} \frac{(-1)^m (2m) \cdot x^{2m-1}}{2^{2m+n} \cdot m! \cdot (m+n)!}$$

$$= \sum_{m=0}^{\infty} \frac{(-1)^m \cdot x^{2m-1}}{2^{2m+n-1} \cdot (m-1)! \cdot (m+n)!}$$

$$= \sum_{m=0}^{\infty} \frac{(-1)^m \cdot x^{2m-1+n}}{x^{n+1} \cdot 2^{2m+n-1} \cdot (m-1)! \cdot (m+n)!}$$

$$= x^{-n} \sum_{m=0}^{\infty} \frac{(-1)^m \cdot x^{2m-1+n}}{2^{2m+n-1} \cdot (m-1)! \cdot (m+n)!}$$

Put $m-1 = s$,
 $m = s+1$

$$= x^{-n} \sum_{s=0}^{\infty} \frac{(-1)^{s+1} \cdot x^{2(s+1)-1+n}}{2^{2(s+1)+n-1} \cdot s! \cdot (s+1+n)!}$$

$$= -x^{-n} \sum_{s=0}^{\infty} \frac{(-1)^s \cdot x^{2s+2-1+n}}{2^{2s+2+n-1} \cdot s! \cdot (s+n+1)!}$$

$$= -x^{-n} \sum_{s=0}^{\infty} \frac{(-1)^s}{2^{2s+n+1}} x^{2s+n+1} \cdot s! \cdot (s+n+1)!$$

$$= -x^{-n} J_{n+1}(x)$$

$$= \text{R.H.S. (proved)}$$

Q. no. 3 (a)

$$i) L \left\{ \frac{e^t \sin at}{t} \right\}$$

Solution:

$$L \{ \sin at \} = \frac{a}{s^2 + a^2}$$

$$L \{ e^{-t} \sin at \} = \left[\frac{a}{s^2 + a^2} \right]_{s \rightarrow s+1}$$

$$= \frac{a}{(s+1)^2 + a^2}$$

$$= \frac{a}{s^2 + 2s + 1 + a^2}$$

Now,

$$L \{ e^{-t} \sin at \} = \int_0^{\infty} e^{-st} \sin at \, dt$$

$$= \int_0^{\infty} e^{-st} \sin at \, dt$$

P

$$= \frac{a}{s^2 + 2s + 1 + a^2}$$

+1

$(s+n+1)!$

Now,

$$L \left\{ \frac{e^{-t} \sin t}{t} \right\}$$

$$= \int_s^\infty \left[\frac{a}{(s+1)^2 + a^2} \right] ds$$

Put $s+1 = x$,
 $ds = dx$,

$$= \int_{x=1}^\infty \left[\frac{a}{x^2 + a^2} \right] dx$$

$$= a \left[\frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) \right]_{x=1}^\infty$$

$$= a \left[\tan^{-1} \left(\frac{s+1}{a} \right) \right]_{s=1}^\infty$$

$$= a \left[\tan^{-1} \infty - \tan^{-1} \left(\frac{s+1}{a} \right) \right]$$

$$= a \left[\frac{\pi}{2} - \tan^{-1} \left(\frac{s+1}{a} \right) \right]$$

$$\text{ii) } L^{-1} \left\{ \frac{1}{(s-1)(s^2+9)} \right\}$$

solution;

$$\text{Let } F(s) = \frac{1}{s-1}$$

$$f(t) = L^{-1} \{ F(s) \} = e^t$$

$$G(s) = \frac{1}{s^2+9}$$

$$= \frac{1}{3} \sin 3t$$

$$g(t) = L^{-1} \{ G(s) \} = \frac{1}{3} \sin 3t$$

Q

Now, by using convolution,

$$f(t) * g(t) = e^t * \frac{1}{3} \sin 3t$$

~~Question~~

Q. no. 5 (a)

Green's theorem states that the integral $\oint_C [f(x,y) dx + g(x,y) dy]$

can be evaluated as,

$$\iint_S \left[\frac{\partial}{\partial x} g(x,y) - \frac{\partial}{\partial y} f(x,y) \right] dA$$

To calculate :

$$\oint_C [(y - \sin x) dx + \cos x dy]$$

$$= \iint_S \left[\frac{\partial}{\partial x} (\cos x) - \frac{\partial}{\partial y} (y - \sin x) \right] dA$$

$$= \iint_S [-\sin x - 1] dA$$

Q. no. 3(b)

Given, $y'' - 2y' + y = e^t$ — ①

$$\begin{aligned} y(0) &= 2 \\ y'(0) &= -1 \end{aligned}$$

Taking Laplace transform on both of ①,

$$L\{y'' - 2y' + y\} = L\{e^t\}$$

$$\text{or, } L\{y''\} - 2L\{y'\} + L\{y\} = \frac{1}{s-1}$$

$$\text{or, } s^2 L(y) - s y(0) - y'(0) - 2(s L(y) - y(0)) + L(y) = \frac{1}{s-1}$$

$$\text{or, } s^2 L(y) - 2s + 1 - 2s L(y) + 4 - 2L(y) = \frac{1}{s-1}$$

$$\text{or, } L(y) \{s^2 - 2s - 2\} - 2s + 1 + 4 = \frac{1}{s-1}$$

$$\text{or, } L(y) \{s^2 - 2s - 2\} = \frac{1}{s-1} + 2s - 5$$

$$\text{or, } L(y) \{s^2 - 2s - 2\} = \frac{1 + (s-1)(2s-5)}{(s-1)}$$

$$\text{or, } L\{y\} =$$

$$\text{or, } L\{y\} =$$

Let

2

$$\text{or } L\{y\} = \frac{1+2s^2-5s-2s+5}{(s-1)(s^2-2s-2)}$$

$$\text{or } L\{y\} = \frac{2s^2-7s+6}{(s-1)(s^2-2s-2)}$$

Let

2

both

$$\frac{1}{-1}$$

$$= \frac{1}{s-1}$$

$$4 = \frac{1}{s-1}$$

$$2s-5$$

$$(2s-5)$$

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ional answerbook No. : 144

ber of additional answerbooks used :

e of the student : Prashant Bhat

No. (Class) : 54 Exam : Preba

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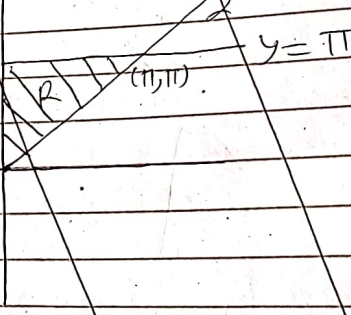
is answerbook must be stiched properly

Q. no. 1(a)

Let $I = \int_0^{\pi} \int_x^{\pi} \frac{\sin y}{y} dy dx$

$\cos 2x = 1 - 2\sin^2 x$
 $\sin^2 x = \frac{1 - \cos 2x}{2}$

$1 - \cos 2x$



$\frac{d \cos \pi y}{d \pi y} \cdot \frac{d \pi y}{dy}$

$= -\sin \pi y$

$\frac{1}{\pi} \frac{d \sin \pi y}{dy} \cdot \frac{d \pi y}{dy}$
 $= \cos \pi y \cdot \pi$

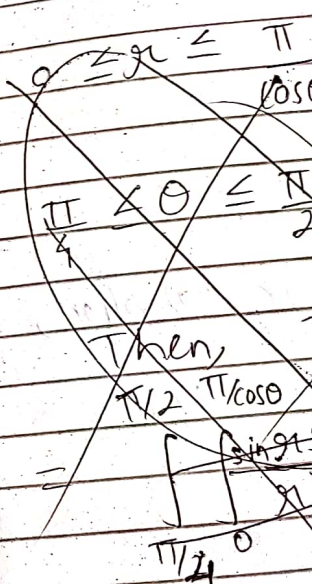
Put $x = r \cos \theta$
 $y = r \sin \theta$

When $x = 0$, $r \cos \theta = 0 \Rightarrow r = 0$

$x = \pi$, $r \cos \theta = \pi \Rightarrow r = \frac{\pi}{\cos \theta}$

When, $y = x$, $r \sin \theta = r \cos \theta$
 $\tan \theta = 1$
 $\theta = \frac{\pi}{4}$

When, $y = \pi$, $r \sin \theta = \pi$



$I =$



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Number of additional answerbooks used :
Name of the student : Prashant Bhatta
Roll No. (Class) : 54
Exam : Preboard
Grade/Semester/ : III
Programme : B.E. Computer
Exam :
Date : 2080-10-24
Inviigilator's Signature :

Students are required to write answer on both sides starting from the first page of this copy.
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$$0 \leq \theta \leq \pi$$

$$\pi/2 \leq \theta \leq \pi$$

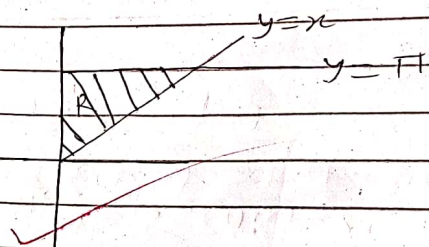
Then, I

$$= \int_{\pi/2}^{\pi} \int_0^{\pi \cos \theta} x \sin \theta \, dx \, d\theta$$

$$I = \int_0^{\pi} \int_0^y \frac{\sin y}{y} \, dx \, dy$$

Q.no. 1 (a)

$$\text{Let } I = \int_0^{\pi} \int_x^{\pi} \frac{\sin y}{y} dy dx$$



Changing the order of integration,

x varies from 0 to $x=y$.
 y varies from 0 to π .

Then,

$$I = \int_0^{\pi} \int_0^y \frac{\sin y}{y} dx dy$$

$$= \int_0^{\pi} \frac{\sin y}{y} [x]_0^y dy$$

$$= \int_0^{\pi} \frac{\sin y \cdot y}{y} dy$$

$$= \int_0^{\pi} \sin y dy$$

$$= [-\cos y]_0^{\pi}$$

$$= -[\cos \pi - \cos 0]$$

$$= -[-1 - 1]$$

$$= -(-2)$$

$$= 2$$

7

Q. no. 1 (b)

$$V: x \geq 0, y \geq 0, z \geq 0$$

$$\text{and } \frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1, \quad \text{--- (1)}$$

$$\text{Put } \frac{x}{a} = u,$$

$$x = au$$

$$dx = a du$$

$$\text{Put } \frac{y}{b} = v$$

$$y = bv$$

$$dy = b dv$$

$$\text{Put } \frac{z}{c} = w$$

$$z = cw$$

$$dz = c dw$$

Then, ① becomes,

$$u + v + w = 1.$$

Now,

$$\iiint_V x^2 dx dy dz$$

$$= \iiint_V (aw)^2 \cdot a du \cdot b dv \cdot c dw$$

$$= \iiint_V a^2 u^2 \cdot a b c du dv dw$$

$$= \iiint_V a^3$$

$$= a^3 b c$$

$$= a^3 b c$$

$$= a^3 b c$$

$$= \frac{2 a^3}{5}$$

$$= \frac{2 a^3}{12}$$

$$= \frac{a^3 b}{60}$$

Theore
Dirich
in
in

then
as,

$$= \iiint_V a^3 b c u^2 du dv dw$$

$$= a^3 b c \iiint_V u^{3-1} \cdot v^{1-1} \cdot w^{1-1} du dv dw$$

$$= a^3 b c \cdot \frac{\Gamma 3 \Gamma 1 \Gamma 1}{\Gamma 3+1+1+1} \quad [\because \text{from Dirichlet's theorem}]$$

$$= a^3 b c \cdot \frac{2!}{\Gamma 6}$$

$$= \frac{2 a^3 b c}{5!}$$

$$= \frac{2 a^3 b c}{120}$$

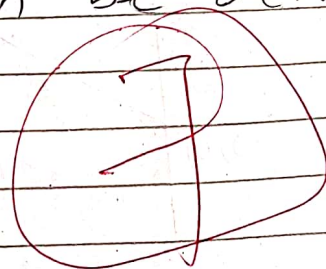
$$= \frac{a^3 b c}{60}$$

Theorem:

Dirichlet's theorem states that if ~~integ~~ $(x+y+z=1)$ and integral is in the form $\iiint_V x^{l-1} y^{m-1} z^{n-1} dv$

then, it's volume can be defined as,

$$\frac{\Gamma l \Gamma m \Gamma n}{\Gamma l+m+n+1}$$



Q. no. 5 (a)

$$\oint_C [(y - \sin x) dx + \cos x dy],$$

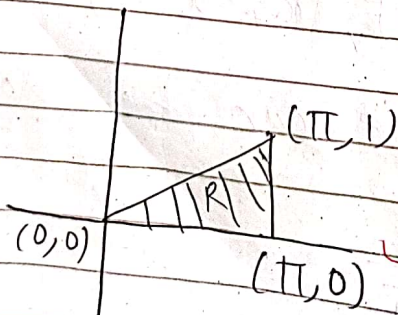
C is the triangle with vertices $(0, 0)$, $(\pi, 0)$, $(\pi, 1)$

To calculate:

$$\oint_C [(y - \sin x) dx + \cos x dy]$$

$$= \iint_S \left[\frac{\partial \cos x}{\partial x} + \frac{\partial (y - \sin x)}{\partial y} \right] dA$$

$$= \iint_S [-\sin x - 1] dA$$



~~varies~~
 x varies from 0 to π

$$y - 0 = \frac{1 - 0}{\pi - 0} (x - 0)$$

$$\pi y = x$$

x varies from 0 to πy .

y varies from 0 to π .

Then, $\pi \times \pi$

~~$$\int_0^{\pi} \int_0^{\pi} [-\sin x - 1] dx dy$$~~

$$\int_0^{\pi} \int_0^{\pi y} [-\sin x - 1] dx dy$$

$$= \int_0^{\pi} [\cos x - x]_0^{\pi y} dy$$

$$= \int_0^{\pi} [\cos \pi y - \cos 0 - \pi y] dy$$

$$= \int_0^{\pi} [\cos \pi y - 1 - \pi y] dy$$

$$= \left[\frac{1}{\pi} \sin \pi y - y - \frac{\pi y^2}{2} \right]_0^{\pi}$$

$$= \frac{\sin \pi^2}{\pi} - \frac{\sin 0}{\pi} - \pi - \frac{\pi \cdot \pi^2}{2}$$

$$= 0 - 0 - \pi - \frac{\pi^3}{2}$$

$$= -\pi - \frac{\pi^3}{2}$$

Theorem:

Green's theorem states that, the integral $\oint_C [f(x, y) dx + g(x, y) dy]$

can be evaluated as

$$\iint_S \left[\frac{\partial g(x, y)}{\partial x} - \frac{\partial f(x, y)}{\partial y} \right] dA$$



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No. (Class) : 34

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National Academy of science and Technology

[Affiliated to Pokhara University]

Dhangadhi, Kailali (Nepal)

Accredited by University Grants Commission (UGC), Nepal (2022)

Additional answerbook No. :
Number of additional answerbooks used : 1442 Invigilator's Signature : 
Name of the student : Prashant Bhalta Programme : BE Computer
Roll No. (Class) : 34 Exam : Pre board Grade/Semester/ : III Exam :
Paper : Date : 2080-10-24

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Q. no. 6 (b)

Partial differential equation is,
 $u_t + 2u \cdot u_x = 0$ — (1)

It's auxiliary equation is,

$$\frac{dt}{1} = \frac{dx}{2u} = \frac{du}{0}$$

Taking first and second,

$$\frac{dt}{1} = \frac{dx}{2u}$$

$$\text{or } 2u \int dt = \int dx$$

$$\text{or } 2u \cdot t = x + C_1$$

$$\text{or } C_1 = 2ut - x$$

Taking second and third,

$$\frac{dx}{2u} = \frac{du}{0}$$

$$\text{or } 0 = 2u du$$

$$\text{or } 2 \int u du + C_2 = 0$$

$$\text{or } 2 \frac{u^2}{2} + C_2 = 0$$

$$\text{or } C_2 = -u^2$$

$$\text{or } f(u) = -u^2$$

$$u(x, t) = \text{cancel} - u^2$$

$$u(x, 0) = \text{cancel} - u^2$$

Q. no. 7 (a)

~~$e^{-3t} u(t-2)$~~
 ~~$e^{-3(t-2+2)} u(t-2)$~~
 ~~$e^{-3(t-2)-6} u(t-2)$~~

i) $e^{-3t} \cdot u(t-2)$

$$= e^{-3(t-2+2)} \cdot u(t-2)$$

$$= e^{-3(t-2)-6} \cdot u(t-2)$$

$$= e^{-6} \cdot e^{-3(t-2)} \cdot u(t-2)$$

$$\mathcal{L}^{-1} \{ e^{-6} \cdot e^{-3(t-2)} \cdot u(t-2) \}$$

$$= e^{-6} \cdot e^{-6s} \cdot u_2(t)$$

$$(V) \int_0^{\pi} \int_0^{\sin x} y \, dy \, dx$$

$$= \int_0^{\pi} \left[\frac{y^2}{2} \right]_0^{\sin x} dx$$

$$= \int_0^{\pi} \frac{\sin^2 x}{2} dx$$

$$= \int_0^{\pi} \frac{1 - \cos 2x}{4} dx$$

$$= \frac{1}{4} \int_0^{\pi} [1 - \cos 2x] dx$$

$$= \frac{1}{4} \left[x - \frac{\cos 2x}{2} \right]_0^{\pi}$$

$$= \frac{1}{4} \left[\pi - \frac{\cos 2\pi}{2} + \frac{\cos 0}{2} \right]$$

$$= \frac{1}{4} \left[\pi - \frac{1}{2} + \frac{1}{2} \right]$$

$$= \frac{\pi}{4}$$

ii)

$$f =$$

$$\vec{r} = \nabla f$$

At

$$\vec{r} =$$

$$\hat{n} =$$

$$= \frac{\pi}{4}$$

ii)

$$f = x^2 y + 2xz - 4$$

$$\vec{r} = \nabla f = \hat{i} \frac{\partial f}{\partial x} + \hat{j} \frac{\partial f}{\partial y} + \hat{k} \frac{\partial f}{\partial z}$$

$$= (2xy + 2z)\hat{i} + x^2\hat{j} + 2x\hat{k}$$

$$\text{At } (2, -2, 3) = (x, y, z).$$

$$\vec{r} = (-8 + 6)\hat{i} + 4\hat{j} + 4\hat{k}$$

$$= -2\hat{i} + 4\hat{j} + 4\hat{k}$$

Unit normal vector is given by,

$$\hat{n} = \frac{\vec{r}}{|\vec{r}|}$$

$$= \frac{-2\hat{i} + 4\hat{j} + 4\hat{k}}{\sqrt{4 + 16 + 16}}$$

$$= \frac{-2\hat{i} + 4\hat{j} + 4\hat{k}}{6}$$

$$= -\frac{1}{3}(\hat{i} - 2\hat{j} - 2\hat{k})$$

$$= -\frac{1}{3}(\hat{i} - 2\hat{j} - 2\hat{k})$$

2.5

$$\text{iv) } f(x) = \begin{cases} \frac{1}{2} + x & \text{for } -\frac{1}{2} < x < 0 \\ \frac{1}{2} - x & \text{for } 0 < x < \frac{1}{2} \end{cases}$$

$$\text{For } -\frac{1}{2} < x < 0,$$

$$f(x) = \frac{1}{2} + x$$

$$f(-x) = \frac{1}{2} - x$$

which is a ~~odd~~ odd function.

~~So we need to check further,~~
the

$$\text{For } 0 < x < \frac{1}{2},$$

$$f(x) = \frac{1}{2} - x$$

$$f(-x) = \frac{1}{2} - (-x)$$

$$= \frac{1}{2} + x$$

It means neither odd nor even.

$$-\frac{1}{2} < x < 0$$

$$0 < x < \frac{1}{2}$$

∴ the series is neither odd nor even.

$$\text{Period} = \frac{1}{2} - \left(-\frac{1}{2}\right)$$

$$= \frac{1}{2} + \frac{1}{2}$$

$$= 1$$

Q. no. 6 (a)

Given, $f(x) = x^2$, $-\pi \leq x \leq \pi$.

The fourier series is given by

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \quad \text{--- (1)}$$

where,

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx$$

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} x^2 dx$$

$$= \frac{1}{2\pi} \cdot 2 \int_0^{\pi} x^2 dx \quad [\because \text{Even function}]$$

$$= \frac{1}{\pi} \left[\frac{x^3}{3} \right]_0^{\pi}$$

$$= \frac{\pi^3}{3\pi}$$

$$= \frac{\pi^2}{3}$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 \cos nx dx$$

$$= \frac{2}{\pi} \int_0^{\pi} x^2 \cos nx dx$$

[$\because$ Even function]



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Additional answerbook
Number of additional an

Name of the student : P2

Roll No. (Class) : 34

Paper :

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$$= \frac{2}{\pi}$$

$$= \frac{2}{\pi}$$

$$= \frac{2}{\pi}$$

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$$= \frac{2}{\pi}$$

$$= \frac{2}{\pi}$$



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National answerbook No. : 1470

Number of additional answerbooks used : 1

Name of the student : Prashant Bhatta

Invigilator's Signature : [Signature]

Roll No. (Class) : 34

Programme : BE Computer

Exam : Preboard

Grade/Semester/

Exam :

Paper :

Date : 2080-10-24

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f: Even function

$$= \frac{2}{\pi} \left[x^2 \left(\frac{\sin nx}{n} \right) - 2x \left(\frac{-\cos nx}{n^2} \right) \right]$$

$$= \frac{2}{\pi} \left[x^2 \left(\frac{\sin nx}{n} \right) - 2x \left(\frac{-\cos nx}{n^2} \right) + 2 \left(\frac{-\sin nx}{n^3} \right) \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[2\pi \cdot \frac{\cos n\pi}{n^2} \right]$$

$$= \frac{4}{n^2} \cos n\pi$$

bn = 1

Then, from ①,

$$f(x) = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \frac{4}{n^2} \cos n\pi \cdot \cos nx$$

5

Put $x=0$,

$$f(0) = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \frac{4}{n^2} \cos n\pi \cdot \cos n\pi$$

$$\text{or, } 0 = \frac{\pi^2}{3} +$$

Q.no. 4(a)

Theorem:

The necessary and sufficient condition for a vector function $\vec{a} = \vec{a}(t)$ to have constant direction is

$$\vec{a} \times \frac{d\vec{a}}{dt} = 0.$$

Necessary Condition:

Let $\vec{a}$ has constant direction.
We have to prove:

$$\vec{a} \times \frac{d\vec{a}}{dt} = 0$$

$$\vec{a} = a \hat{a}$$

Differentiating both sides w.r. to t

$$\frac{d\vec{a}}{dt} = \frac{d(a \hat{a})}{dt}$$

$$\text{or } \frac{d\vec{a}}{dt} = a \frac{d\hat{a}}{dt} + \hat{a} \frac{da}{dt}$$

$$\text{or } \frac{d\vec{a}}{dt} = 0 + \hat{a} \frac{da}{dt}$$

(cross multiplying by $\vec{a}$ on both sides,

$$\begin{aligned} \vec{a} \times \frac{d\vec{a}}{dt} &= \vec{a} \times \hat{a} \frac{da}{dt} \\ &= \cancel{\vec{a} \times a \hat{a}} \times \hat{a} \frac{da}{dt} \\ &= a (\hat{a} \times \hat{a}) \frac{da}{dt} \\ &= 0 \end{aligned}$$

$$\text{Hence, } \vec{a} \times \frac{d\vec{a}}{dt} = 0$$

Sufficient Condition:

$$\text{Let } \vec{a} \times \frac{d\vec{a}}{dt} = 0$$

We have to prove $\vec{a}$ has constant direction,

$$\vec{a} = a \hat{a}$$

Differentiating both sides w.r. to t

$$\frac{d\vec{a}}{dt} = \frac{d(a \hat{a})}{dt}$$

$$\frac{d\vec{a}}{dt} = a \frac{d\hat{a}}{dt} + \hat{a} \frac{da}{dt}$$

$$\frac{d\vec{a}}{dt} - a \frac{d\hat{a}}{dt} = 0$$

Cross multiplying by $\vec{a}$ on both sides,

$$\vec{a} \times \frac{d\vec{a}}{dt} = \vec{a} \times a \frac{d\hat{a}}{dt}$$

$$\text{or } 0 = a \hat{a} \times a \frac{d\hat{a}}{dt}$$

$$\text{or } 0 = a^2 \hat{a} \frac{d\hat{a}}{dt}$$

$$\cancel{\vec{a} \cdot \frac{d\vec{a}}{dt} = 0} \text{ or, } 0 = a \vec{a} \frac{d\vec{a}}{dt}$$

Also,

$$\vec{a} \cdot \frac{d\vec{a}}{dt} = 0$$

Since,

$$\vec{a} \neq 0.$$

$$\text{so, } \frac{d\vec{a}}{dt} = 0$$

$\Rightarrow \vec{a}$ has constant direction.

(Proved)

Q. no. 4 (b)

$$(x, y, z) = (t^3 - 4t, t^2 + 4t, 8t^2 - 3t^3)$$

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$= (t^3 - 4t)\hat{i} + (t^2 + 4t)\hat{j} + (8t^2 - 3t^3)\hat{k}$$

$$x = t^3 - 4t$$

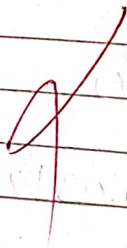
$$dx = (3t^2 - 4) dt$$

$$y = t^2 + 4t$$

$$dy = (2t + 4) dt$$

$$z = 8t^2 - 3t^3$$

$$dz = (16t - 9t^2) dt$$



Q.no. 5 (b)

$$\vec{F} = x^2 \hat{i} - e^y \hat{j} + \hat{k}$$

$$S: x + y + z - 1 = 0$$

$$\begin{aligned}\nabla S &= \hat{i} \frac{\partial S}{\partial x} + \hat{j} \frac{\partial S}{\partial y} + \hat{k} \frac{\partial S}{\partial z} \\ &= \hat{i} + \hat{j} + \hat{k}\end{aligned}$$

$$\begin{aligned}|\nabla S| &= \sqrt{1^2 + 1^2 + 1^2} \\ &= \sqrt{3}\end{aligned}$$

$$\hat{n} = \frac{1}{\sqrt{3}} (\hat{i} + \hat{j} + \hat{k})$$